Salmon and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Protacanthopterygii
ORDERS	3
FAMILIES	16
SPECIES	538

During their lifetime, many species of salmon and trout make impressive journeys from the oceans (where they live as adults) to freshwater rivers (where they spawn). Salmon are well known for their spectacular leaps up waterfalls as they make their way upriver. Freshwater pikes are close relatives of salmon. Salmon, trout, and pike are native to North America, Europe, and Asia, although the former two have been widely introduced elsewhere. Three other groups of fish are also considered relatives: freshwater galaxiids, marine smelts or argentinoids (which include tubeshoulders and spookfish), and freshwater smelts or osmeriforms (such as capelin and eulachon). However, recent work now suggests these 3 groups may be more closely related to dragonfish and in future may be placed with them in a new superorder called Osmeromorpha.

Anatomy

Salmon and their relatives have a slim, tapering body, usually with well-developed swimming muscles. The fins are relatively small, except for the tail fin, which is usually large and powerful, enabling the fishes to swim and maneuver quickly. Most species have a small, fleshy fin, known as an adipose fin, near the base of the tail, and pelvic fins that are located far back on the body. Scales in these fishes are either small or entirely absent. As carnivores that feed on a wide variety of prey, most species have a large mouth studded with many sharp teeth. The highly predatory pikes have long teeth for seizing prey and an especially large mouth that allows them to swallow prey almost half their own size. Salmon, trout, and pikes often ambush prey or attack with a burst of speed over a short distance.

Most galaxiids are small fishes, less than 10 in (25 cm) long, with a tubular, scaleless body. They have a square tail and, unlike most other fishes in this group, no adipose fin.

Some of the marine fishes in this group have unusual adaptations for life in deep water. The argentinoids, for example, include some species with large, tubular eyes that point forward or upward (giving good binocular vision). Other argentinoids have light-emitting organs.

Life cycle

Some species in this group complete their entire life in either the sea or in freshwater. However, other species, including many salmon and trout, spend most of their life in the ocean but move to freshwater to breed, a life cycle described as anadromous. Males and females spawn in freshwater rivers. When the young hatch, they grow for a short time in freshwater and then move out to sea. After a period that varies from a few months to several years, the fishes mature into adults and return to freshwater to spawn, usually to the stream where they hatched. To complete the journey, which may be several thousand miles long, they often have to negotiate fast currents or rough terrain, requiring them to leap clear of the water to progress upstream. Breeding adults usually cease feeding before beginning their migration. Having used all their energy on the journey, they usually die soon after spawning. How they find their way to their hatching site is not entirely understood, but the sense of smell and perhaps celestial navigation are thought to be important. Not all salmon and trout follow this type of life cycle: those belonging to landlocked populations spend all their life in freshwater.

Some species of galaxiids are also anadromous. The tiny larvae hatch in freshwater and are swept out to sea. After a few months, they return as juveniles to freshwater, where they mature into adults.

BREEDING MALES
During the migration to their breeding grounds, the males of many species of salmon undergo drastic physical changes. These include the appearance of new colors and markings. Some species also develop a hump on their back and elongated hooked jaws. These pink salmon from the Pacific Ocean are swimming up a stream in Alaska.